

§3 Emergent Geometry

The geometric layer of the Reconciliation Causal Framework, rewritten for v2.1 with the SINGLE LARGEST STRUCTURAL CHANGE in the rewrite: **5 GAPS FLAGGED**

Study metric swap ($d_{FS} = \ell_0 \cdot \arccos(\sqrt{K_\omega})$ replaces $-\ell_0 \cdot \log K_\omega$). Triangle inequality becomes automatic; Factorization + Dominance apparatus removed. Introduces $K_\omega J$ (Jordan-built kernel, fixes F2). $D=3$ from Gram rank (representation- theoretic argument, GAP-3.6.5 FATAL). Log retained as potential function (Hessian gives metric tensor). Five inline gap flags surface the representation-theoretic question and K_ω vs $K_\omega J$ coexistence.

10
SUBSECTIONS

C
LAYER

10+
DEFINITIONS

6
THEOREMS

VERSION
AUDIENCE
DATE

v2.1 Draft (Canonical Form)
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RCF Section 3 — Emergent Geometry

Rewritten Canonical Form — v2.1 (Draft)

Preamble — What This Section Contains and Why

Section 0 established the algebraic substrate (A, Lie–Jordan decomposition, admissibility, GNS, algebraic F, two co-primitives). Section 1 produced the dynamical foundation (Open Expansion, fracture, burden, Tier 1/Tier 2 split, A_red, M_red, R_t, Convergence Theorem, thin/full A_phy, events, causal order $\prec$). Section 2 assembled the QM layer (Reconciliation Principle, dark energy mechanism, fields, particles, mass, records, Born rule, FIREWALL). With the microscopic dynamics closed, Section 3 constructs the emergent geometry — the operational "where" built from the Jordan structure (locality primitive, §0.4.2) and the GNS state.

What changed from v2.0

v2.0's §3 was "largely intact in content" relative to v1.0's §2, with four targeted corrections (D=3 circularity fix, phase-preserving kernel in the volume element, Type-Sign T-4 dual-use resolved, smooth field convergence). v2.1 makes the **single largest structural change in the entire rewrite**: the Fubini-Study metric swap. The changes are:

1. **Fubini-Study metric adoption** (§3.2, NEW — the biggest change). v2.0 used $d_\omega = -\square_0 \log K_\omega$ (log distance), which is NOT an unconditional metric — it requires Cubic Factorization (Condition 3.3.1) AND Cubic Dominance (Condition 3.3.2) to force the triangle inequality. v2.1 swaps to $d_{FS} = \square_0 \cdot (\sqrt{K_\omega})$ — the Fubini-Study geodesic distance on projective Hilbert space. The triangle inequality becomes AUTOMATIC (geodesic distances are metrics). This kills the Factorization + Dominance apparatus outright.

2. **Log retained as potential function** (§3.2.4, NEW). The $-\log(K_\omega)$ is NOT discarded — it is reinterpreted as a potential function whose Hessian gives the Fubini-Study metric tensor. This is the resolution of the user's "–log vs arccos" question: use both, in different roles. The metric (arccos) handles geometry; the potential (log) handles information theory. Standard in Riemannian geometry and Petz 1996 quantum information geometry.

3. **K_ω^J Jordan-built kernel** (§3.1.2, NEW — fixes F2 from patch plan). v2.0's $K_\omega(A, B) = |\omega_{\text{kin}}(A^\dagger B)|^2 / (\cdot s)$ does NOT use the Jordan product $A \circ B$ — despite Remark 3.1.3's claim that the symmetry "comes from the Jordan structure." v2.1 introduces the Jordan-built variant: $K_\omega^J(A, B) = \omega_{\text{kin}}(A \circ B) / \sqrt{\omega_{\text{kin}}(A^\dagger A) \cdot \omega_{\text{kin}}(B^\dagger B)}$. This genuinely uses $A \circ B = 12(AB + BA)$. The Fubini-Study distance $d_{FS}^J = \square_0 \cdot (|K_\omega^J|)$ is still a metric (arccos on any inner-product space).

4. **D=3 from Gram rank** (§3.6, EXPANDED). v2.0 demoted the D=3 closure from "Theorem" to "Conditional Proposition" because the closure defect $_C(D)$ was tautological ($|D-3|$). v2.1 keeps the demotion but reframes the argument: the Gram matrix of K_ω^J has rank ≤ 3 because the Jordan algebra A_J supports exactly three independent symmetric bilinear invariants. The closure defect $_C(D) =$

(D-3)₊ measures excess rank — non-tautological, with rank-3 forced by the algebra's representation theory.

5. **Cubic kernel as Riemannian volume element** (§3.6.2, REINTERPRETED). v2.0's corrected volume element $V_\omega = |\det K_\omega|$ (using the phase-preserving kernel) is reinterpreted as the Riemannian volume element of the Fubini-Study metric on the 3-dimensional submanifold spanned by $[A], [B], [C] \in P(\square_{\text{kin}})$. Standard differential geometry (Gram determinant = volume²).

6. **Factorization and Dominance removed** (§3.3, REMOVED). v2.0's Conditions 3.3.1 (Cubic Factorization) and 3.3.2 (Cubic Dominance) were introduced solely to force the triangle inequality for the log distance. Under the Fubini-Study metric, these conditions are superfluous for metricity. The product structure $K_\omega(A, C) = K_\omega(A, B) \cdot K_\omega(B, C)$ (Factorization) remains a property of the KERNEL K_ω (not the distance) and may still be used where needed downstream.

7. **Π_{net} relocated here** (§3.9, retained from v2.0). The cross-extension network operator, forward-referenced in v1.0 §0.3.8, is defined here after K_ω and the sectoral decomposition are both available.

8. **Smooth field convergence** (§3.8, retained from v2.0). Introduced as a complementary mechanism to the triangle inequality for the coarse-graining bridge. Under the Fubini-Study metric, the triangle inequality is automatic, but smooth field convergence remains needed for the continuum limit (the metric gives metricity; the field convergence gives smoothness).

9. **Orientation Invariant dropped** (retained from v2.0). v1.0's quarantined three-body chirality invariant 0_ω is dropped with no replacement.

What this section does NOT contain (deferred)

- Duration, burden-clock $\alpha(B)$, the "now" ($\rightarrow$ §4)
- The burden tensor $^{\wedge}(B)_{\mu\nu}$, Einstein closure, gravity ($\rightarrow$ §4)
- Black holes, cosmology ($\rightarrow$ §5, §6)

The test for this section: does it contain anything that depends on duration, the burden tensor, gravity, or cosmology? If yes, the reordering has failed. (Exception: §3.9's Π_{net} is used by §4.4's relational burden channel, but Π_{net} is defined here purely algebraically — its gravitational application is deferred.)

§3.0 Purpose: The Operational "Where"

Section 0 named locality as a structural co-primitive, grounded in the Jordan algebra A_J (Definition 0.4.2). Section 1 activated locality via Open Expansion, which drives local Jordan-built sectors to reconcile with the global causal Lie layer. The fracture (§1.2) decomposed $\square_{\text{kin}}$ into sub-sectors — the locality sectors. With the thin/full A_{phy} split (§1.7) and zero-preserving events (§1.8) available, we can now construct the operational "where": the emergent spatial geometry.

The key sequence is:

1. **Localisable observables** A_{loc} (§3.1) — the sector where the correlation kernel is well-defined.

2. **Correlation kernel** K_ω and the **Jordan-built variant** K_ω^J (§3.1) — the algebraic seeds of geometry.
 3. **Fubini-Study metric** d_{FS} (§3.2) — the geodesic distance on projective GNS space. Triangle inequality automatic.
 4. **Log potential** $\Phi = -\log K_\omega$ (§3.2) — the information-theoretic potential whose Hessian gives the metric tensor.
 5. **Metric quotient** (X_ω, d_{FS}) (§3.4) — the emergent space of correlation-equivalence classes.
 6. **Direction** (§3.5) — the tangent space, now a genuine Riemannian object.
 7. **D=3 closure** (§3.6) — from the Gram rank of the Jordan algebra's three independent invariants.
 8. **Type-Sign Coupling** (§3.7) — necessary condition for Lorentzian signature; sufficiency deferred to §4.
 9. **Coarse-graining bridge** (§3.8) — smooth field convergence to the continuum.
 10. **Cross-extension network operator** Π^{net} (§3.9) — the network topology, now definable after K_ω .
- Every object is defined before use; the chain is acyclic. The closing Acyclicity Test (§3.10) verifies this.

§3.1 The Correlation Kernel and the Jordan-Built Variant

Definition 3.1.1 (Localisable Observables)

[Retained from v2.0, with the v2.0 fix: A_{loc} is fully determined, not chosen.]

The **localisable observable sector** is:

$$A_{\text{loc}} := \{A \in A_{\text{phy}}^{\text{thin}} : \omega_{\text{kin}}(A^\dagger A) > 0\}$$

This is the set of thin-physical observables (§1.7.1) with non-vanishing GNS norm. The definition is fully determined — every element of $A_{\text{phy}}^{\text{thin}}$ with positive GNS norm is included, no selection involved. We require $1 \in A_{\text{loc}}$.

Definition 3.1.2 (Correlation Kernel — v2.0 sesquilinear form)

[Retained from v2.0 as the "standard" kernel. v2.1 introduces the Jordan-built variant K_ω^J in Definition 3.1.3.]

The **correlation kernel** (sesquilinear form) is:

$$K_\omega(A, B) := \frac{|\omega_{\text{kin}}(A^\dagger B)|^2}{\omega_{\text{kin}}(A^\dagger A) \cdot \omega_{\text{kin}}(B^\dagger B)}$$

This is the squared cosine of the angle between $[A]_\omega$ and $[B]_\omega$ in the GNS Hilbert space. It is symmetric ($K_\omega(A, B) = K_\omega(B, A)$), positive, and bounded.

Definition 3.1.3 (Jordan-Built Correlation Kernel — NEW in v2.1)

[NEW in v2.1. This fixes $F2$ from the patch plan: v2.0's K_ω does not use the Jordan product $A \circ B$, despite Remark 3.1.3's claim.]

The **Jordan-built correlation kernel** is:

$$K_\omega^J(A, B) := \frac{\omega_{\text{kin}}(A \circ B)}{\sqrt{\omega_{\text{kin}}(A^\dagger A) \cdot \omega_{\text{kin}}(B^\dagger B)}}$$

where $A \circ B = \frac{1}{2}(AB + BA)$ is the Jordan product (Definition 0.1.2).

Properties:

1. **Uses the Jordan product:** K_{ω^J} genuinely uses $A \circ B$, unlike K_{ω} which uses the raw sesquilinear form $\omega_{\text{kin}}(A^\dagger B)$.
2. **Symmetric by construction:** $K_{\omega^J}(A, B) = K_{\omega^J}(B, A)$ because $A \circ B = B \circ A$.
3. **Bounded:** $|K_{\omega^J}(A, B)| \leq 1$ (by the Cauchy-Schwarz inequality applied to the Jordan product in the GNS representation).
4. **Real-valued:** $\omega_{\text{kin}}(A \circ B) \in \mathbb{R}$ (because $A \circ B$ is self-adjoint when A, B are self-adjoint, and ω_{kin} is a state).

Remark 3.1.4 (K_{ω} vs. K_{ω^J} — which is the correct kernel?)

[This is the resolution of GAP-0.3.4 from §0.]

v2.0's K_{ω} uses the raw sesquilinear form $\omega_{\text{kin}}(A^\dagger B)$. Its symmetry comes from state-Hermiticity ($|\omega(A^\dagger B)| = |\omega(B^\dagger A)|$), NOT from the Jordan structure — despite v2.0's Remark 3.1.3 claiming otherwise. The framework's two-primitive picture says locality = Jordan (§0.4.2), so the principled choice is K_{ω^J} , which genuinely uses the Jordan product.

v2.1 adopts K_{ω^J} as the primary kernel for geometry. The downstream formulas (Fubini-Study metric, cubic kernel, $D=3$ closure) are re-derived with K_{ω^J} . The standard K_{ω} (Definition 3.1.2) is retained for backward compatibility and for contexts where the sesquilinear form is more natural (e.g., the Born rule's Fubini-Study distance on pure states, §2.7.5, which uses $|\langle \psi | \phi \rangle|^2 = K_{\omega}(|\psi\rangle, |\phi\rangle)$).

[GAP: v2.1 — whether K_{ω^J} is the correct kernel for ALL downstream geometry, or only for the spatial geometry (while K_{ω} is correct for the Born rule), is not yet fully resolved. The question is: does the Born rule (§2.7.5) use K_{ω} (sesquilinear, natural for pure states) or K_{ω^J} (Jordan-built, natural for spatial geometry)? The Phase C program (Ch 4, L7) suggests both can coexist — K_{ω} for the Born rule's projective geometry, K_{ω^J} for the spatial geometry. This coexistence needs verification. Flagged as GAP-3.1.4, severity MATERIAL (affects the unity of the geometric picture).]

Definition 3.1.5 (Phase-Preserving Kernel)

[Retained from v2.0, with K_{ω^J} variant noted.]

The **phase-preserving kernel** is:

$$\tilde{K}_{\omega}(A, B) := \frac{\omega_{\text{kin}}(A^\dagger B)}{\sqrt{\omega_{\text{kin}}(A^\dagger A) \omega_{\text{kin}}(B^\dagger B)}}$$

where $\phi_{AB} = \omega_{\text{kin}}(A^\dagger B)$ is the relative phase. The phase-preserving kernel retains the complex phase information that K_{ω} (magnitude-only) discards.

The **Jordan-built phase-preserving variant** is:

$$\tilde{K}_{\omega^J}(A, B) := \frac{\omega_{\text{kin}}(A \circ B)}{\sqrt{\omega_{\text{kin}}(A \circ A) \omega_{\text{kin}}(B \circ B)}}$$

which is real-valued (no phase to preserve — the Jordan product is symmetric, so the "phase" is always 0 or π). This is a key difference: K_{ω^J} is real, while K_{ω} is complex.

Remark 3.1.6 (Phase-preserving kernel and the Gram determinant)

The v2.0 correction (using K_ω instead of K_ω^J in the volume element, §3.6.2) was motivated by the Gram determinant's need for complex phase information to detect linear dependence. With K_ω^J (real-valued), the Gram determinant is automatically real — the question is whether a real Gram determinant is sufficient for the $D=3$ closure argument, or whether the complex phase structure of K_ω is essential.

[GAP: v2.1 — the interplay between K_ω^J (real, Jordan-built) and K_ω (complex, sesquilinear) in the $D=3$ closure argument needs clarification. The v2.0 correction used K_ω (complex) for the Gram determinant. If v2.1 adopts K_ω^J (real) as the primary kernel, does the Gram determinant still detect linear dependence correctly? A real Gram determinant detects real linear dependence but may miss complex linear dependence. This is a technical question about the representation theory of Jordan algebras over $\mathbb{C}$ vs. $\mathbb{R}$. Flagged as GAP-3.1.6, severity MATERIAL (affects the $D=3$ closure proof).]

Remark 3.1.7 (K_ω and the Jordan structure — honest statement)

[This replaces v2.0's Remark 3.1.3, which incorrectly claimed the symmetry "comes from the Jordan structure."]

v2.0's Remark 3.1.3 claimed: "The symmetry of K_ω comes from the Jordan structure." This was incorrect — the proof v2.0 gave ($|\omega(A^\dagger B)| = |\omega(B^\dagger A)|$) uses only state-Hermiticity, never touching $A \circ B$.

v2.1 honest statement: K_ω 's symmetry comes from state-Hermiticity (the GNS inner product is Hermitian). The Jordan structure licenses the INTERPRETATION of K_ω as a correlation (symmetric, undirected — a "where"-type quantity), but does not GENERATE the symmetry. The Jordan-built variant K_ω^J (Definition 3.1.3) is the kernel whose symmetry genuinely comes from the Jordan product. This is the F2 fix from the patch plan, applied.

§3.2 The Fubini-Study Metric and the Log Potential

Definition 3.2.1 (Fubini-Study Distance — NEW in v2.1)

[NEW in v2.1. This is the single largest structural change in the entire rewrite. v2.0 used $d_\omega = -\square_0 \log K_\omega$ (log distance); v2.1 swaps to the Fubini-Study geodesic distance.]

The **Fubini-Study distance** between two observables $A, B \in A_{\text{loc}}$ is:

$$d_{\text{FS}}(A, B) := \ell_0 \cdot \arccos(\sqrt{K_\omega(A, B)}) = \ell_0 \cdot \arccos$$

This is the geodesic distance on projective GNS space $P(\square_{\text{kin}})$ equipped with the Fubini-Study metric tensor. It takes values in $[0, \square_0 \cdot \pi/2]$.

The **Jordan-built variant** is:

$$d_{\text{FS}}^J(A, B) := \ell_0 \cdot \arccos(|K_\omega^J(A, B)|)$$

Both are metrics on $P(\square_{\text{kin}})$ (Lemma 3.2.2).

Lemma 3.2.2 (Fubini-Study is a Metric)

[This is L1 from the Phase C program deep-dive appendix.]

The Fubini-Study distance d_{FS} (and its Jordan-built variant d_{FS}^J) is a metric on projective GNS space $P(\square_{\text{kin}})$:

1. **Non-negativity:** $d_{\text{FS}}(A, B) \geq 0$, with equality iff $[A] = [B]$.

2. **Symmetry:** $d_{\text{FS}}(A, B) = d_{\text{FS}}(B, A)$.

3. **Triangle inequality:** $d_{\text{FS}}(A, C) \leq d_{\text{FS}}(A, B) + d_{\text{FS}}(B, C)$ — AUTOMATIC (geodesic distance on a Riemannian manifold).

Proof. Standard. d_{FS} is the geodesic distance on $P(\square_{\text{kin}})$ with the Fubini-Study metric tensor. Geodesic distances on Riemannian manifolds satisfy the triangle inequality by definition (the geodesic from A to C is no longer than the path through B). Non-negativity and symmetry are immediate. The Jordan-built variant d_{FS}^J is also a metric because K_{ω}^J is a valid inner-product magnitude (by the Cauchy-Schwarz inequality), and arccos of any inner-product magnitude is a metric on the projective space.

Theorem 3.2.3 (Triangle Inequality is Automatic — KILLS Factorization + Dominance)

[This is L2–L3 from the Phase C program deep-dive appendix. The biggest single consequence of the Fubini-Study swap.]

For any three admissible observables $A, B, C \in A_{\text{loc}}$, the Fubini-Study distance satisfies:

$$d_{\text{FS}}(A, C) \leq d_{\text{FS}}(A, B) + d_{\text{FS}}(B, C)$$

without any Factorization or Dominance condition.

Proof. Direct corollary of Lemma 3.2.2. The geodesic distance on any Riemannian manifold satisfies the triangle inequality. $P(\square_{\text{kin}})$ with the Fubini-Study metric is a Riemannian manifold. Therefore $d_{\text{FS}}(A, C) \leq d_{\text{FS}}(A, B) + d_{\text{FS}}(B, C)$ for all A, B, C — no conditions on the triad needed.

Remark 3.2.4 (What this kills). v2.0's Conditions 3.3.1 (Cubic Factorization: $K_{\omega}(A, C) \approx K_{\omega}(A, B) \cdot K_{\omega}(B, C)$) and 3.3.2 (Cubic Dominance: $K_{\omega}(A, C) \geq K_{\omega}(A, B, C)$) were introduced SOLELY to force the triangle inequality for the log distance $d_{\omega} = -\square_0 \log K_{\omega}$. Under d_{FS} , these conditions are superfluous for metricity. They are REMOVED from v2.1's §3.3.

The product structure $K_{\omega}(A, C) = K_{\omega}(A, B) \cdot K_{\omega}(B, C)$ (Factorization, when it holds) remains a property of the KERNEL K_{ω} — not of the distance. It may still be used where needed downstream (e.g., in the cubic kernel's multiplicative structure), but it is no longer a condition for the triangle inequality.

Definition 3.2.5 (Log Potential — RETAINED, REINTERPRETED)

[NEW interpretation in v2.1. The $-\log K_{\omega}$ is NOT discarded — it is reinterpreted as a potential function.]

The **log potential** is:

$$\Phi(A, B) := -\log K_{\omega}(A, B) = -\log |\tilde{K}_{\omega}(A, B)|^2 = -2 \log |\tilde{K}_{\omega}|$$

This is the Rényi-2 relative entropy (also called the "logarithmic distance" or "potential of the metric"). It is NOT a competitor to the Fubini-Study distance — it is the POTENTIAL whose Hessian gives the metric tensor.

Lemma 3.2.6 (Log Potential and Fubini-Study Metric Tensor)

[This is L4 from the Phase C program deep-dive appendix. The resolution of the user's "–log vs arccos" question.]

The Fubini-Study metric tensor g^{FS}_{ij} is the Hessian of the log potential:

$$g^{\text{FS}}_{ij} = \frac{1}{2} \cdot \frac{\partial^2 \Phi}{\partial \theta^i \partial \theta^j} \Big|_{\text{diag}}$$

where θ^i are local coordinates on $P(\Pi_{\text{kin}})$.

Proof sketch. Standard quantum information geometry. The Fubini-Study metric tensor on $P(\Pi_{\text{kin}})$ in local coordinates is $g^{\text{FS}}_{ij} = \frac{1}{2} \frac{\partial^2}{\partial \theta^i \partial \theta^j} (-\log |\langle \psi | \psi \rangle|^2)$ evaluated at the diagonal. Equivalently, g^{FS} is the Hessian of the Rényi-2 relative entropy potential $-\log K_\omega$. This is the standard relationship between a Riemannian metric and its potential function (analogous to $g_{ij} = \frac{\partial^2 \phi}{\partial x^i \partial x^j}$ in Hessian geometry).

Remark 3.2.7 (The resolution of the $-\log$ vs $\arccos$ question). The user's concern — "replacing one with the other will just be changing the problem" — is exactly right IF we try to use one function for both purposes. The resolution is to stop trying:

- **Fubini-Study ($\arccos$)** is the METRIC — the geodesic distance. It gives the triangle inequality, the tangent space, the geodesic structure, the Riemannian volume element. It is the primary geometric object.

- **Log potential ($-\log K_\omega$)** is the POTENTIAL FUNCTION — the information-theoretic content. It appears in κ_B , BH entropy, and the cubic kernel's multiplicative structure. Its Hessian gives the metric tensor.

The two are related by DIFFERENTIATION, not by substitution. The metric is the Hessian of the potential — this is standard Riemannian geometry. Neither replaces the other.

Remark 3.2.8 (κ_B and BH entropy survive the swap)

[This is L6 from the Phase C program deep-dive appendix. The ripple-effect guarantee.]

The formulas $\kappa_B = C/(\Pi_{\text{max}} \cdot \Pi_0^2)$ and $S_{\partial R} = \text{Area}(\partial R)/(4\Pi_0^2)$ use Π_0 as the fundamental scale from the ALGEBRA's spectral discreteness (the Π_0 -floor, §5.1.3: $\Pi_{\text{max}} = C\Pi/\Pi_0^2$), NOT from the distance function. Switching d_ω from $-\Pi_0 \log K_\omega$ to $d_{\text{FS}} = \Pi_0 (\sqrt{K_\omega})$ keeps the same Π_0 prefactor (the scale doesn't change) — only the SHAPE of the distance function changes ($\log \rightarrow \sqrt{}$). The κ_B and BH formulas depend on Π_0 as a SCALE, not on the shape of d_ω , so they are unaffected.

§3.3 Metricity (SIMPLIFIED — Factorization and Dominance Removed)

Theorem 3.3.1 (Pseudometric — AUTOMATIC)

[v2.1: SIMPLIFIED. v2.0 had Conditions 3.3.1 (Factorization) and 3.3.2 (Dominance) plus Theorem 3.3.3 (Pseudometric, conditional). v2.1 replaces all three with a single automatic theorem.]

The Fubini-Study distance d_{FS} is a pseudometric on A_{loc} : it satisfies non-negativity, symmetry, and the triangle inequality (Theorem 3.2.3), with $d_{\text{FS}}(A, A) = 0$.

Proof. Direct from Lemma 3.2.2 and Theorem 3.2.3. No conditions on the kernel or the triad are needed.

Remark 3.3.2 (What was removed). v2.0's §3.3 contained:

- Condition 3.3.1 (Cubic Factorization): $K_\omega(A, C) \approx K_\omega(A, B) \cdot K_\omega(B, C)$ for admissible triads.
- Condition 3.3.2 (Cubic Dominance): $K_\omega(A, C) \geq K_\omega(A, B, C)$.

- Theorem 3.3.3 (Pseudometric, conditional on 3.3.1 + 3.3.2).
- Assumption 3.3.5 (Approximate Metricity, T-2 conditional).

v2.1 removes ALL FOUR. The triangle inequality is automatic under the Fubini-Study metric (Theorem 3.2.3). The product structure (Factorization) survives as a property of the kernel K_ω (not the distance) and may be used where needed, but it is no longer a condition for metricity.

[GAP: v2.1 — none in §3.3. The Fubini-Study swap eliminates the entire apparatus of conditional metricity. This is the single biggest simplification in v2.1.]

§3.4 The Metric Quotient

Definition 3.4.1 (Correlation-Equivalence Classes)

Two observables $A, B \in A_{\text{loc}}$ are **correlation-equivalent** if:

$$A \sim_\omega B \quad \Longleftrightarrow \quad d_{\text{FS}}(A, B) = 0 \quad \Longleftrightarrow$$

This means $[A]_\omega$ and $[B]_\omega$ are parallel in the GNS Hilbert space (they represent the same projective point).

Definition 3.4.2 (Emergent Space)

The **emergent space** is the quotient:

$$X_\omega := A_{\text{loc}} / \sim_\omega$$

equipped with the metric $d_{\text{FS}}([A], [B]) := d_{\text{FS}}(A, B)$. The quotient is well-defined because d_{FS} is constant on equivalence classes (it depends only on the projective points).

Theorem 3.4.3 (Metric Space)

$(X_\omega, d_{\text{FS}})$ is a metric space.

Proof. The Fubini-Study distance d_{FS} is a pseudometric on A_{loc} (Theorem 3.3.1) that vanishes exactly on correlation-equivalent pairs (Definition 3.4.1). The quotient of a pseudometric by its kernel is a metric space — standard.

§3.5 Direction and the Tangent Space

Definition 3.5.1 (Displacement Profile)

The **displacement profile** of B relative to A is the function:

$$\Delta_{\{A \rightarrow B\}}(C) := d_{\text{FS}}(B, C) - d_{\text{FS}}(A, C)$$

for all $C \in A_{\text{loc}}$. This measures how the distance to C changes when moving from A to B .

Definition 3.5.2 (Direction as Tangent Vector)

[v2.1: REINTERPRETED. v2.0 treated direction as a profile difference. v2.1, under the Fubini-Study metric, treats direction as a genuine tangent vector on the Riemannian manifold $P(\square_{\text{kin}})$.]

The **direction** from A to B is the tangent vector at [A] on $P(\square_{\text{kin}})$:

$$\vec{\Delta}_{A \rightarrow B} := \left[\frac{d}{dt} \bigg|_{t=0} \exp_{[A]}(t \cdot v_{A \rightarrow B}) \right]$$

where $\exp_{[A]}$ is the Riemannian exponential map at [A] and $v_{A \rightarrow B}$ is the initial velocity of the geodesic from [A] to [B].

Operationally, $\Delta_{A \rightarrow B}$ is the equivalence class of displacement profiles $\Delta_{A \rightarrow B}(\cdot)$ under the scaling relation $\Delta \sim \lambda \Delta$ for $\lambda > 0$ (Def 3.5.3 in v2.0, retained).

Remark 3.5.3 (Direction is now a genuine tangent vector)

Under v2.0's log distance, direction was an ad hoc profile difference — a function $\Delta_{A \rightarrow B}: A_{\text{loc}} \rightarrow \mathbb{R}$, not a geometric object. Under v2.1's Fubini-Study metric, direction is a genuine tangent vector on the Riemannian manifold $P(\square_{\text{kin}})$. This is a significant improvement: the tangent space $T_{[A]} P(\square_{\text{kin}})$ is now a real vector space (not just a space of profiles), and standard Riemannian geometry applies.

Definition 3.5.4 (Emergent Tangent Cone)

The **emergent tangent cone** at $[A] \in X_\omega$ is:

$$T^{\text{em}}_{[A]} X_\omega := \text{span}\{\vec{\Delta}_{A \rightarrow B} : B \in A_{\text{loc}}\}$$

This is the space of all directions from [A] — the tangent space to X_ω at [A].

§3.6 D=3 Closure from Gram Rank

Definition 3.6.1 (Three Inference Channels)

[Retained from v2.0, with the layer mapping clarified.]

The three **spatial inference channels** are:

1. **Existence** (Quadratic, Layer Q): $E(A) := \omega_{\text{kin}}(A^\dagger A) > 0$. An observable exists if it has non-zero GNS norm.
2. **Distance** (Cubic, Layer C): $D(A, B) := d_{\text{FS}}(A, B)$. The Fubini-Study geodesic distance.
3. **Direction** (Cubic, Layer C): $D_{A \rightarrow B} := [\Delta_{A \rightarrow B}] \in T_{[A]} X_\omega$. The tangent vector (§3.5.2).

Remark 3.6.2 (Duration is NOT a spatial channel). Duration (the operational "when") is constructed in §4.1 from burden-weighted causal depth. It is the Lie primitive (§0.4.1) made operational, NOT a fourth spatial channel. The three spatial channels come from the Jordan structure (co-occurrence); the duration channel comes from the Lie structure (ordering). Their complementarity is Theorem 0.4.4.

Definition 3.6.3 (Closure Defect — non-tautological)

[Retained from v2.0. The v1.0 tautological definition ${}_C(D) = |D - 3|$ is replaced by a non-tautological rank-deficiency definition.]

The **closure defect** at dimension D is:

$$\chi_C(D) := \text{rank deficiency of the } D \times D \text{ cubic Gram matrix} + (D - 3)_+$$

where:

- The Gram matrix is $G_{ij} = K_\omega(X_i, X_j)$ (or $G_{ij} = K_\omega^J(X_i, X_j)$ for the Jordan-built variant) for D points $X_1, \dots, X_D \in A_{\text{loc}}$.
- "Rank deficiency" is $D - \text{rank}(G)$ (the number of linearly dependent rows/columns).
- $(D - 3)_+ = \max(D - 3, 0)$ is the excess dimensionality above 3.

This is NOT defined as $|D - 3|$ (the v1.0 tautological definition). It measures (i) the rank deficiency of the Gram matrix (a genuine linear-algebraic quantity) and (ii) the excess dimensionality above 3.

Theorem 3.6.4 (D=3 Closure — CONDITIONAL PROPOSITION)

[Retained from v2.0 as a Conditional Proposition (demoted from v1.0's "Theorem"). v2.1 keeps the demotion but reframes the argument via the Jordan algebra representation theory.]

The closure defect vanishes ($\chi_C(D) = 0$) if and only if $D = 3$, conditional on the Jordan algebra A_J supporting exactly three independent symmetric bilinear invariants.

Status: [Conditional Proposition — v1.0 "Theorem" demoted]. The claim $\chi_C(D) = 0 \Leftrightarrow D = 3$ relies on two structural assumptions:

1. **Three-channel assumption:** existence, distance, and direction are the only spatial inference channels (Definition 3.6.1).
2. **Gram non-degeneracy:** the Gram matrix $G_{ij} = K_\omega(X_i, X_j)$ (or $K_\omega^J(X_i, X_j)$) is non-degenerate for generic admissible sets.

Proof sketch (v2.1 reframe).

Step 1: Gram rank ≤ 3 from Jordan algebra structure. The Gram matrix $G_{ij} = K_\omega^J(X_i, X_j)$ is built from the Jordan product $X_i \circ X_j$. If the Jordan algebra A_J supports exactly three independent symmetric bilinear invariants (existence, distance, direction — the three inference channels), then $\text{rank}(G) \leq 3$ for any set of points, with equality for generic admissible sets.

Step 2: $\chi_C(D) = 0$ iff $D \leq 3$. For $D \leq 3$, the Gram matrix can have full rank ($\text{rank} = D$), so the rank deficiency is 0 and $(D-3)_+ = 0$, giving $\chi_C(D) = 0$. For $D > 3$, the rank is at most 3 (Step 1), so the rank deficiency is $D - 3 \geq 1$ and $(D-3)_+ = D - 3$, giving $\chi_C(D) \geq 2(D-3) > 0$.

Step 3: $D = 3$ is uniquely locked. $D = 1$ (only existence) and $D = 2$ (existence + distance, no direction) do not support the full geometric structure (no tangent space for $D = 2$). $D = 3$ (existence + distance + direction) is the smallest dimension that supports all three channels. $D > 3$ has $\chi_C(D) > 0$ (Step 2). Therefore $D = 3$ is uniquely locked — conditional on the three-channel assumption and the Gram non-degeneracy.

Remark 3.6.5 (The representation-theoretic question — FLAGGED)

[This is GAP-0.1.5, now at its primary location. The hardest open question in the $D=3$ closure argument.]

The $D=3$ closure (Theorem 3.6.4) depends on the claim that the Jordan algebra A_J supports exactly three independent symmetric bilinear invariants. This is a representation-theoretic question:

- The Jordan–von Neumann–Wigner classification (1934) gives the structure theory for finite-dimensional Jordan algebras.
- The framework's A may be infinite-dimensional, and the invariant-counting argument needs development for the infinite-dimensional case.
- The three invariants (existence, distance, direction) are NAMED but not proven to be the ONLY independent symmetric bilinear invariants.

[GAP: v2.1 — the representation-theoretic claim " A_J supports exactly 3 independent symmetric bilinear invariants" is NOT proven. This is the F5 fix from the patch plan (severity FATAL). The $D=3$ closure is honestly demoted to "Conditional Proposition" pending this proof. If the claim fails (e.g., if A_J supports 4 or more independent invariants), the framework predicts extra spatial dimensions or a modified geometry — which would be a major structural problem. Flagged as GAP-3.6.5, severity FATAL (blocks $D=3$ closure, which blocks the dimensional-suppression argument for BH entropy in §5).]

Definition 3.6.6 (Cubic Volume Element — REINTERPRETED)

[Retained from v2.0 (corrected to use K_ω), REINTERPRETED in v2.1 as the Riemannian volume element.]

The **cubic volume element** for three observables $A, B, C \in A_{\text{loc}}$ is:

$$V_\omega(A, B, C) := |\det[\tilde{K}_\omega(X_i, X_j)]_{i,j \in \{A,B,C\}}|$$

v2.1 reinterpretation (L5 from the Phase C program): This is the Riemannian volume element of the Fubini-Study metric on the 3-dimensional submanifold of $P(\square_{\text{kin}})$ spanned by $[A], [B], [C]$. The Gram determinant $|\det K_\omega|$ is the squared volume of the parallelepiped spanned by these three projective points — standard differential geometry.

Remark 3.6.7 (Why the phase-preserving kernel is needed). The v2.0 correction (using K_ω instead of K_ω) is retained. The Gram determinant's ability to detect linear dependence requires the complex phase information that K_ω preserves and K_ω (magnitude-only) discards. The v2.0 internal contradiction (§2.1.2 warned of "96% false-positive rate" for magnitude-only Gram determinants, yet §2.7.2 used exactly that kernel) is resolved.

[GAP: v2.1 — if $K_\omega^\wedge$ (real-valued) is adopted as the primary kernel, the Gram determinant becomes real, and the question is whether a real Gram determinant is sufficient for the $D=3$ closure. See GAP-3.1.6. Flagged as GAP-3.6.7, severity MATERIAL (affects the $D=3$ closure proof's technical details).]

§3.7 Type-Sign Coupling

Theorem 3.7.1 (Algebraic Type Distinction)

The scalar sector A_{scal} (observables that commute with all of A_{loc}) and the vector sector A_{vec} (observables that don't) are algebraically distinct:

$$A_{\text{loc}} = A_{\text{scal}} \oplus A_{\text{vec}}$$

This is a structural fact about the GNS representation.

Theorem 3.7.2 (Type-Sign Coupling — Necessary Condition)

[Retained from v2.0. The T-4 dual-use bug is resolved: T-4 is reserved for the Born rule (§2.7); Lorentzian-signature sufficiency is Theorem 4.5.3.]

The Lorentzian signature $(-, +, +, +)$ is structurally forced as a **necessary condition**: the scalar sector (time) carries the negative sign, and the vector sector (space) carries the positive signs. This follows from the algebraic type distinction (Theorem 3.7.1) and the burden-clock suppression structure (§4.1.1).

Sufficiency (that $(-, +, +, +)$ is the UNIQUE compatible signature) is deferred to §4.5.3 (Theorem 4.5.3, after the full metric is available). This resolves the v1.0/v2.0 "T-4 dual-use bug": T-4 is reserved exclusively for the Born rule (closed/strengthened in §2.7); the Lorentzian-signature sufficiency is now Theorem 4.5.3, an Established result (conditional on T-2).

Definition 3.7.3 (Three-Layer Bridge)

[Retained from v2.0, with the Fubini-Study metric noted.]

The **three-layer bridge** C_ϵ connects the discrete correlation structure to the smooth manifold via coarse-graining:

$$C_\epsilon: (X_\omega, \tilde{d}_{FS}) \rightarrow (M, g)$$

where (M, g) is the smooth Riemannian manifold and ϵ is the coarse-graining scale. The bridge has four effects: (i) metricity (now automatic from d_{FS}), (ii) smoothness (from smooth field convergence, §3.8), (iii) dimensionality ($D = 3$ from §3.6), (iv) signature $(-, +, +, +)$ from §3.7 + §4.5.3).

§3.8 Coarse-Graining and Smooth Field Convergence

Conjecture 3.8.1 (Smooth Field Convergence)

[Retained from v2.0. Under the Fubini-Study metric, the triangle inequality is automatic, but smooth field convergence is still needed for the continuum limit.]

As the density of stable field modes (§2.4) on (X_ω, d_{FS}) increases, the discrete correlation structure converges to a smooth field configuration:

$$\text{As } N_{\text{modes}} \rightarrow \infty, \quad (X_\omega, \tilde{d}_{FS}) \rightarrow (M, g)$$

The smooth field IS the smooth manifold. The continuum limit is achieved not by abstract averaging but by the physical convergence of stable field modes.

Status: [Structural conjecture, T-2/T-3 conditional]. The convergence requires the stable-mode density to increase without bound, which depends on the spectral gap (T-2) and the continuum limit (T-3).

Remark 3.8.2 (Two complementary mechanisms for the continuum limit)

Under v2.1's Fubini-Study metric, the coarse-graining bridge has TWO complementary mechanisms:

1. **Metricity** (automatic from d_{FS} , Theorem 3.2.3) — the triangle inequality.
2. **Smoothness** (from smooth field convergence, Conjecture 3.8.1) — the continuum limit.

v2.0 had a third mechanism (triangle inequality from Factorization + Dominance), which v2.1 removes. The remaining two mechanisms are sufficient: metricity gives the metric space structure; smooth field

convergence gives the smooth manifold.

[GAP: v2.1 — the smooth field convergence conjecture (Conjecture 3.8.1) is T-2/T-3-conditional. The convergence proof requires the spectral gap (T-2) and the continuum-limit existence (T-3). Flagged as GAP-3.8.1, severity MATERIAL (propagated).]

§3.9 The Cross-Extension Network Operator

Definition 3.9.1 (Cross-Extension Network Operator)

[Retained from v2.0. Relocated from v1.0 §0.3.8 — now defined after K_ω and the sectoral decomposition are both available.]

The **cross-extension network operator** is:

$$\hat{\Pi}_{\text{net}} := \sum_{E_i} \hat{P}_{E_i} \cdot K_\omega \cdot \hat{P}_{E_i}$$

where $\hat{P}_{E_i}$ are projectors onto the open-extension sectors (§1.2.3) and K_ω is the correlation kernel (§3.1.2). This operator encodes the network topology of the fractured sector graph — it projects onto the subspace of observables correlated across multiple open-extension sectors.

Remark 3.9.2 (Π_{net} is derivable from $\mathbf{F}$ and K_ω — zero new primitives). Π_{net} is built from $\hat{F}$ (via the sectoral decomposition, §1.2.3) and K_ω (§3.1.2). It introduces NO new primitives — it is a derived object that encodes the network structure of the fractured algebra.

Remark 3.9.3 (Connection to relational burden). The relational burden channel (§4.4.1) is $\hat{(\text{rel})} = [\hat{C}_\alpha, \hat{\Pi}_{\text{net}}]$. This is the network-weighted sum of cross-sector commutators:

$$[\hat{C}_\alpha, \hat{\Pi}_{\text{net}}] = \sum_{i \neq j} K_\omega(i, j) \cdot [\hat{C}_\alpha, P_{E_i} \cdot P_{E_j}]$$

connecting $\hat{F}_{\text{cross}}$ (Theorem 1.2.4) to Π_{net} (this section). The identification $B_{\text{rel}} = \text{Tr}(\rho \cdot \hat{F}_{\text{cross}}) = \sum_{i \neq j} K_\omega(i, j) \cdot B_{\text{cross}}(i, j)$ is formalized in §4.4.

§3.10 Architectural Summary

What Section 3 Contains

Subsection	Content	Layer
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§3.0	Purpose: the operational "where"	—
§3.1	Localisable observables, K_ω , K_ω^J (Jordan-built), phase-preserving kernel	C
§3.2	Fubini-Study metric d_{FS} , log potential, metric tensor	C
§3.3	Metricity (SIMPLIFIED — Factorization + Dominance removed)	C

Subsection	Content	Layer
§3.4	Metric quotient $(X_\omega, \tilde{d}_{\{FS\}})$	C
§3.5	Direction as tangent vector (genuine Riemannian)	C
§3.6	D=3 closure from Gram rank (Conditional Proposition), volume element	C
§3.7	Type-Sign coupling (necessary), three-layer bridge	$C \rightarrow Q'$
§3.8	Smooth field convergence (T-2/T-3 conditional)	$C \rightarrow Q'$
§3.9	$\hat{\Pi}_{\{net\}}$ (cross-extension network operator)	C

What Section 3 Does NOT Contain (Deferred)

Object	Deferred to	Reason
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Burden tensor $\Theta^{\{B\}}_{\{\mu\nu\}}$	§4.3	Requires the emergent metric (§3.4, now available)
Three-tier burden decomposition (as gravitational source)	§4.4	Requires $\hat{\Pi}_{\{net\}}$ (§3.9, now available) and $\hat{F}_{\{cross\}}$ (§1.2.4, available)
Einstein closure, gravity	§4.5	Requires burden tensor (§4.3)
Duration, burden-clock $\alpha(B)$, the "now"	§4.1, §4.2	Requires space (§3, now available) and burden (§1.3, available)
Type-Sign sufficiency	§4.5.3	Requires the full metric (§3.4 + §4.1)
Black holes, cosmology	§5, §6	Requires gravity (§4)

Acyclicity Test

Question: Does Section 3 define every object before using it?

Verification:

- §3.1 uses A_{phy}^{thin} (§1.7), ω_{kin} (§0.3), $A \circ B$ (§0.1.2). ✓
- §3.2 uses K_ω (§3.1), $\square_0$ (§0.3.6 — algebraic, not metric-dependent). ✓
- §3.3 uses d_{FS} (§3.2). ✓
- §3.4 uses d_{FS} (§3.2), A_{loc} (§3.1). ✓
- §3.5 uses d_{FS} (§3.2), X_ω (§3.4). ✓
- §3.6 uses K_ω , K_ω , K_ω^J (§3.1), Jordan algebra A_J (§0.1.2). ✓

- §3.7 uses A_{loc} (§3.1), d_{FS} (§3.2). ✓
- §3.8 uses X_{ω} (§3.4), stable field modes (§2.4). ✓
- §3.9 uses P^{E_i} (§1.2.3), K_{ω} (§3.1). ✓

Forward references:

- §3.5.2 references the Riemannian exponential map (standard differential geometry, not a framework object). ✓
- §3.6.4 references the Jordan algebra representation theory (GAP-3.6.5, flagged). ✓
- §3.7.2 references §4.5.3 (Type-Sign sufficiency) — named as deferred, not used. ✓
- §3.8.1 references T-2/T-3 (open targets) — named as conditional, not used. ✓
- §3.9.3 references §4.4.1 (relational burden channel) — named as the downstream consequence, not used. ✓

No forward reference creates a circularity.

Verdict: Section 3 is acyclic. The architecture holds through the emergent geometry.

Inline Gap Summary for §3

Gap ID	Location	Description	Severity	Blocks
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GAP-3.1.4	§3.1 Remark 3.1.4	K_{ω} vs K_{ω}^J coexistence (Born rule vs spatial geometry) not resolved	MATERIAL	Unity of geometric picture
GAP-3.1.6	§3.1 Remark 3.1.6	Real (K_{ω}^J) vs complex ($\tilde{K}_{\omega}$) Gram determinant sufficiency	MATERIAL	D=3 closure technical details
GAP-3.6.5	§3.6 Remark 3.6.5	Representation-theoretic claim " A_J supports exactly 3 invariants" NOT proven	FATAL	D=3 closure, BH entropy (§5)
GAP-3.6.7	§3.6 Remark 3.6.7	K_{ω}^J real Gram determinant sufficiency for D=3	MATERIAL	D=3 closure proof
GAP-3.8.1	§3.8 Conjecture 3.8.1	Smooth field convergence T-2/T-3-conditional	MATERIAL	Continuum limit (propagated)

Total new gaps surfaced in §3: 5 (1 fatal, 4 material). Of these:

- GAP-3.6.5 is GAP-0.1.5 at its primary location — the hardest open question in the framework (representation-theoretic claim for $D=3$).
- GAP-3.1.4 and GAP-3.1.6 are NEW gaps introduced by the v2.1 K_{ω^J} adoption — the question of whether the Jordan-built kernel is the correct choice for ALL downstream geometry, or only for the spatial geometry.
- GAP-3.6.7 is a technical sub-question of GAP-3.1.6 (real vs complex Gram determinant).
- GAP-3.8.1 is a T-2/T-3-propagated dependency.

NEW gaps introduced by v2.1 §3: 2 (GAP-3.1.4, GAP-3.1.6) — both arising from the K_{ω^J} adoption, both MATERIAL, neither FATAL. The Fubini-Study swap itself introduces NO new gaps (it eliminates the Factorization + Dominance apparatus, which was a source of v2.0 complexity).

Section 3 is CLOSED. Section 4 (Time, Gravity, and the Three-Tier Burden) can be merged against this stable geometric foundation.